**2.** One of the terms in the binomial expansion of $(3 + ax)^6$, where $a$ is a constant, is $540x^4$

(a) Find the possible values of $a$.

**(4)**

(b) Hence find the term independent of $x$ in the expansion of

$$\left(\frac{1}{81} + \frac{1}{x^6}\right)(3 + ax)^6$$

**(3)**